KAIZEN MATHS

Lesson starter

Product, Quotient and Chain Rule

Product, Quotient, and Chain Rules

Name _____

S1

Differentiate with respect to x.

$y = (3x + 4)^3(x - 2)^2$

S2

Differentiate with respect to x.

$y = (3x + 5)^4$

S3

Differentiate with respect to x.

$y = \dfrac{5x^2}{2x + 2}$

S4

Differentiate with respect to x.

$y = (x^2 + 3x)^3 \sin(2x)$

Show your working

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ALG-1258-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Product, Quotient and Chain Rule

Product, Quotient, and Chain Rules

Name _____

E1

Differentiate with respect to x.

$y = (2x + 2)^4(x - 3)^2$

E2

Differentiate with respect to x.

$y = (5x + 5)^3$

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Differentiate with respect to x.

$y = \dfrac{3x^4}{4x + 3}$

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Show your working

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Differentiate with respect to x.

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ALG-1258-L2 · E

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Differentiate with respect to x.

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Show your working

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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S1

Differentiate with respect to x.

$y = (3x + 4)^3(x - 2)^2$

Answer:

$\frac{dy}{dx} = 9(3x + 4)^2(x - 2)^2 + (3x + 4)^3 2(x - 2)$

Working:

$u = (3x + 4)^3, \quad v = (x - 2)^2$

$u' = 3(3x + 4)^{3-1} \cdot 3$

$u' = 9(3x + 4)^2$

$v' = 2(x - 2)^{2-1} \cdot 1$

$v' = 2(x - 2)$

$y' = u'v + uv'$

$y' = (9(3x + 4)^2)((x - 2)^2) + ((3x + 4)^3)(2(x - 2))$

S2

Differentiate with respect to x.

$y = (3x + 5)^4$

Answer:

$\frac{dy}{dx} = 12(3x + 5)^3$

Working:

$u = 3x + 5, \quad y = u^4$

$\frac{dy}{du} = 4u^{4-1}$

$\frac{dy}{du} = 4u^3$

$\frac{du}{dx} = 3$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$\frac{dy}{dx} = 4u^3 \cdot 3$

$\frac{dy}{dx} = 4(3x + 5)^3 \cdot 3$

$\frac{dy}{dx} = 12(3x + 5)^3$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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S3

Differentiate with respect to x.

$$y = \frac{5x^2}{2x+2}$$

Answer:

$$\frac{dy}{dx} = \frac{10x^1(2x+2) - (5x^2)(2)}{(2x+2)^2}$$

Working:

$u = 5x^2, \quad v = 2x+2$

$u' = 5 \cdot 2x^{2-1}$

$u' = 10x^1$

$v' = 2$

$y' = \dfrac{u'v - uv'}{v^2}$

$u'v = (10x^1)(2x+2)$

$uv' = (5x^2)(2)$

$v^2 = (2x+2)^2$

$\dfrac{dy}{dx} = \dfrac{10x^1(2x+2) - (5x^2)(2)}{(2x+2)^2}$

S4

Differentiate with respect to x.

$$y = (x^2+3x)^3 \sin(2x)$$

Answer:

$$\frac{dy}{dx} = 3(x^2+3x)^2(2x+3)\sin(2x) + 2(x^2+3x)^3\cos(2x)$$

Working:

$u = (x^2+3x)^3, \quad v = \sin(2x)$

$w = x^2+3x, \quad w' = 2x+3$

$u' = 3w^2w'$

$u' = 3(x^2+3x)^2(2x+3)$

$v' = \cos(2x) \cdot 2$

$v' = 2\cos(2x)$

$y' = u'v + uv'$

$y' = [3(x^2+3x)^2(2x+3)]\sin(2x) + (x^2+3x)^3[2\cos(2x)]$

$\dfrac{dy}{dx} = 3(x^2+3x)^2(2x+3)\sin(2x) + 2(x^2+3x)^3\cos(2x)$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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E1

Differentiate with respect to x.

$y = (2x + 2)^4(x - 3)^2$

Answer:

$\frac{dy}{dx} = 8(2x + 2)^3(x - 3)^2 + (2x + 2)^4 2(x - 3)$

Working:

$u = (2x + 2)^4, \quad v = (x - 3)^2$

$u' = 4(2x + 2)^{4-1} \cdot 2$

$u' = 8(2x + 2)^3$

$v' = 2(x - 3)^{2-1} \cdot 1$

$v' = 2(x - 3)$

$y' = u'v + uv'$

$y' = (8(2x + 2)^3)((x - 3)^2) + ((2x + 2)^4)(2(x - 3))$

E2

Differentiate with respect to x.

$y = (5x + 5)^3$

Answer:

$\frac{dy}{dx} = 15(5x + 5)^2$

Working:

$u = 5x + 5, \quad y = u^3$

$\frac{dy}{du} = 3u^{3-1}$

$\frac{dy}{du} = 3u^2$

$\frac{du}{dx} = 5$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$\frac{dy}{dx} = 3u^2 \cdot 5$

$\frac{dy}{dx} = 3(5x + 5)^2 \cdot 5$

$\frac{dy}{dx} = 15(5x + 5)^2$

Teacher answers & working

Product, Quotient and Chain Rule · Product, Quotient, and Chain Rules

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E3

Differentiate with respect to x.

$$y = \frac{3x^4}{4x+3}$$

Answer:

$$\frac{dy}{dx} = \frac{12x^3(4x+3) - (3x^4)(4)}{(4x+3)^2}$$

Working:

$u = 3x^4, \quad v = 4x+3$

$u' = 3 \cdot 4x^{4-1}$

$u' = 12x^3$

$v' = 4$

$y' = \frac{u'v - uv'}{v^2}$

$u'v = (12x^3)(4x+3)$

$uv' = (3x^4)(4)$

$v^2 = (4x+3)^2$

$\frac{dy}{dx} = \frac{12x^3(4x+3) - (3x^4)(4)}{(4x+3)^2}$